

Zeeshan Manzoor Bhat

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Research Interests

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

Education

Ph.D. in Structural and Earthquake Engineering

July 2018 – March 2025

Indian Institute of Technology Roorkee, India

Thesis: *Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes*

Advisors: [Prof. Yogendra Singh](#), [Prof. Pankaj Agarwal](#)

M.Tech. in Structural Engineering

July 2016 – July 2018

National Institute of Technology Srinagar, India

Thesis: *An Experimental Investigation into the Behavior of Steel-Timber Composite Beams*

Advisor: [Prof. Javed Ahmad Bhat](#)

B.Tech. in Civil Engineering

July 2012 – June 2016

National Institute of Technology Srinagar, India

Research Experience

Research Associate

June 2025 – Present

Indian Institute of Technology Roorkee, India

Project: *Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement*

Funded Projects

Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels

Funding: INR 38 Lakh, NBCC (India) Limited

Institution: Indian Institute of Technology Roorkee, India

Publications

- [1] **Bhat, Z. M.**, & Singh, Y. (2026). Out-of-plane seismic fragility analysis of masonry infills in low- and mid-rise RC frame archetype buildings accounting for in-plane damage. *Journal of Building Engineering*. DOI
- [2] **Bhat, Z. M.**, & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. *Journal of Earthquake Engineering*. DOI
- [3] **Bhat, Z. M.**, & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. *Journal of Structural Engineering*. DOI
- [4] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. *Engineering Structures*. DOI
- [5] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with full-field strain using digital image correlation. *Construction and Building Materials*. DOI
- [6] Waseem, S. A., **Bhat, Z. M.**, & Bhat, J. A. (2022). Experimental investigation into the behavior of steel-timber composite beams. *Practice Periodical on Structural Design and Construction*. DOI

Under Review

- [7] **Bhat, Z. M.**, & Singh, Y. Cyclic behavior and damage assessment of masonry infilled RC frames with openings using DIC. *Engineering Structures*

International Conferences

- [1] **Bhat, Z. M.**, & Singh, Y. (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. In *COMPdyn 2025: Proceedings of the 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*, Rhodes Island, Greece, 15–18 June 2025. [DOI](#)
- [2] **Bhat, Z. M.**, & Singh, Y. (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. In *15th Canadian Masonry Symposium*, Ottawa, Canada, 2–5 June 2025.

Technical Reports

- [1] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels*. NBCC (India) Limited, India.
- [2] Team Member. (2022). *Compendium of Traditional Earthquake-Resilient Construction Practices*. NDMA, Government of India. [Report Link](#)

Open Datasets

- [1] **Bhat, Z. M.**, & Singh, Y. (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. *DesignSafe-CI*. [DOI](#)

Awards and Achievements

- Canada Impact+ Research Training Award, Natural Sciences and Engineering Research Council of Canada (NSERC), 2026
- MHRD Fellowship for Doctoral Studies, Ministry of Human Resource Development, Government of India
- MHRD Fellowship for M.Tech Studies, Ministry of Human Resource Development, Government of India
- Ranked among the top 2% in the Graduate Aptitude Test in Engineering (GATE) 2016, among 119,873 candidates

Technical Skills

- **Programming & Tools:** Python, MATLAB, TCL, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

International Workshops

- **6th International Course on Seismic Analysis of Structures** 19–22 July 2021
University of Palermo, Italy
- **Nonlinear Modelling of RC Structures Using OpenSees** 22–26 June 2020
Università degli Studi di Napoli Federico II, Italy

Teaching Assistantships

- **Advanced Earthquake Resistant Design of Structures (EQN-512)** IIT Roorkee
Instructor: [Prof. Yogendra Singh](#)
- **Earthquake Resistant Design of Structures (EQN-563)** IIT Roorkee
Instructor: [Prof. Yogendra Singh](#)

References

Prof. Yogendra Singh

Professor, Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India

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Prof. Pankaj Agarwal

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Prof. Pradeep Bhargava

Professor, Department of Civil Engineering, Indian Institute of Technology Roorkee, India

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